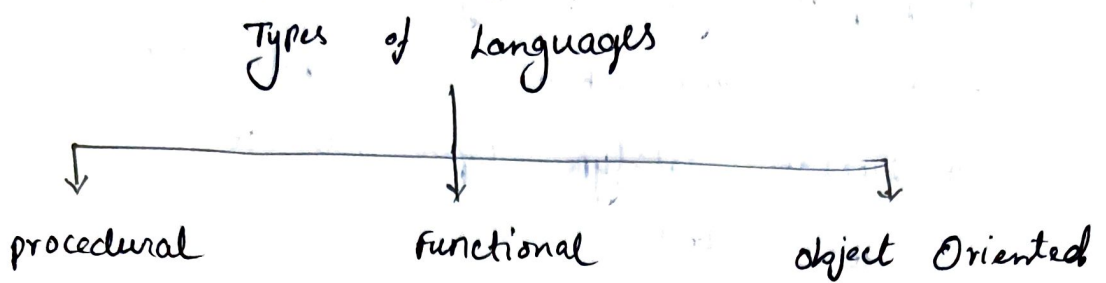


17/07/25

Introduction to programming - Types of Language, memory management.



1) procedural:

- * series of well structured steps and procedures to compose a program.
- * contains a systematic order of statements, functions & commands to complete task.

2) Functional:

- * program only in pure function i.e never modifies variables, but only create new ones as an op.
- * used in situation where we have to perform lots of different operations on the same set of data.

3) Object Oriented:

- * Revolves around objects
- * Code + Data = object

Static Vs Dynamic Languages :-

1) Static :

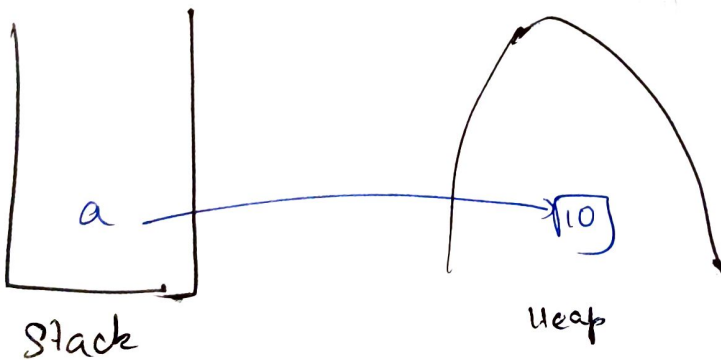
- * perform type checking at compile time
- * Errors will show at compile time
- * Declare datatype before use it
- * More control.

2) Dynamic :

- * perform type checking at runtime.
- * Error might not show till program is run.
- * No need to declare datatypes of variable
- * Saves time in writing code but might give error at runtime.

$a = 10$
↓
ref. variable ↓
 object

(ref var) $a \rightarrow$ stores a reference (address) to where the value 10 is stored in memory. It doesn't actually store the value.



for primitive variables, the variable directly contains the value 10

